Lecture Notes on the IEEE 9-Bus AC Power-Flow Problem

Generated from the loadflow-lite MATPOWER fixtures

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Abstract

The IEEE 9-bus system is a compact benchmark for the steady-state AC power-flow problem. It is small enough to inspect by hand, but it still contains the main modeling features that make power flow nontrivial: multiple generator buses, voltage magnitude control, nonlinear active- and reactive-power balance equations, branch losses, and reactive-power exchange with transmission lines. These notes formulate the problem, describe the specific 9-bus network used in this repository, and discuss the solved operating point recorded in the checked-in MATPOWER-derived fixtures.

1 Context and Relevance

Power-flow analysis is the basic steady-state calculation behind many power-system engineering tasks. Given a network model and specified generation, load, and voltage-control data, the AC power-flow problem computes the complex voltage at each bus and the real and reactive power flows on each branch. The result is not a dynamic trajectory; it is a snapshot of an operating point under balanced, sinusoidal, three-phase conditions.

This calculation matters because many operational and planning decisions depend on quantities that are not specified directly in the input data: voltage magnitudes at load buses, bus voltage angles, slack-bus real-power output, generator reactive-power output, branch flows, and system losses. A candidate dispatch may satisfy a real-power energy balance in aggregate while still producing unacceptable voltage profiles, excessive line flows, or reactive-power requirements. AC power flow is therefore the entry point for contingency analysis, voltage-security assessment, optimal power flow, state estimation validation, and many simulation studies.

The IEEE 9-bus case is useful pedagogically because it is small but not degenerate. It has one slack bus, two PV generator buses, six PQ buses, and nine branches on a 100 MVA base. The topology contains a generator/load network rather than a single radial feeder, and the solution exhibits both real-power losses and significant reactive-power effects from line charging.

2 Mathematical Formulation

2.1 Network Model

Let the system have n buses and complex bus-voltage phasors

$$V_i = |V_i| e^{j\theta_i}, \qquad i = 1, \ldots, n.$$

The nodal admittance matrix is

$$Y_{\text{bus}} = G + jB,$$

where G_{ik} and B_{ik} are the conductance and susceptance entries. At each bus, the net complex power injection is

$$S_i = P_i + jQ_i = V_i I_i^*.$$

Using $I = Y_{\text{bus}} V$, the standard polar-form AC power-flow equations are

$$P_i = |V_i| \sum_{k=1}^{n} |V_k| \left[G_{ik} \cos(\theta_i - \theta_k) + B_{ik} \sin(\theta_i - \theta_k) \right], \tag{1}$$

$$Q_i = |V_i| \sum_{k=1}^{n} |V_k| \left[G_{ik} \sin(\theta_i - \theta_k) - B_{ik} \cos(\theta_i - \theta_k) \right]. \tag{2}$$

The specified net injections are formed from positive generation and positive load:

$$P_i^{\text{spec}} = P_{G,i} - P_{D,i}, \qquad Q_i^{\text{spec}} = Q_{G,i} - Q_{D,i}.$$

All real and reactive powers in these notes are reported on the 100 MVA base used by the fixture files unless explicitly shown in MW or MVAr.

2.2 Bus Types and Unknowns

The power-flow equations are solved with different specifications at different bus types:

- **Slack bus**: specifies $|V|$ and θ. Its real and reactive generation are outputs of the solution. It absorbs the imbalance caused by losses and numerical consistency.
- **PV bus**: specifies real-power generation P_G and voltage magnitude $|V|$. The voltage angle θ and reactive generation Q_G are solved.
- **PQ bus**: specifies real and reactive load, and any fixed generation if present. Both $|V|$ and θ are solved.

For a Newton-Raphson solution, the nonlinear equations are written as mismatches. For non-slack buses,

$$\Delta P_i = P_i^{\text{spec}} - P_i(V, \theta),$$

and for PQ buses,

$$\Delta Q_i = Q_i^{\text{spec}} - Q_i(V, \theta).$$

The Jacobian maps small changes in voltage angles and PQ-bus voltage magnitudes to these active- and reactive-power mismatches. Iteration continues until the mismatches are sufficiently small.

3 The Specific IEEE 9-Bus Network

The fixture used here is `data/case9.json`, with solved values from `data/case9_solution.json`. The case has three generator buses: bus 1 is the slack bus, while buses 2 and 3 are PV buses. Loads are located at buses 5, 7, and 9. The total load is

$$P_D = 315.00 \text{ MW}, \qquad Q_D = 115.00 \text{ MVAr}.$$

The case-side scheduled generation sums to 320.30 MW and 22.62 MVAr, but the solved generator outputs differ because the slack bus and PV-bus reactive outputs are determined by the AC solution.

Table 1: Bus data and solved voltages for the IEEE 9-bus case.

Bus	Type	P_D MW	Q_D MVAr	P_G^{case} MW	Q_G^{case} MVAr	$\lvert V \rvert$ p.u.	θ deg	P_G^{sol} MW
1	slack	0.00	0.00	72.30	27.03	1.0400	0.0000	71.64
2	PV	0.00	0.00	163.00	6.54	1.0250	9.2800	163.00
3	PV	0.00	0.00	85.00	-10.95	1.0250	4.6648	85.00
4	PQ	0.00	0.00	0.00	0.00	1.0258	-2.2168	0.00
5	PQ	90.00	30.00	0.00	0.00	1.0127	-3.6874	0.00
6	PQ	0.00	0.00	0.00	0.00	1.0324	1.9667	0.00
7	PQ	100.00	35.00	0.00	0.00	1.0159	0.7275	0.00
8	PQ	0.00	0.00	0.00	0.00	1.0258	3.7197	0.00
9	PQ	125.00	50.00	0.00	0.00	0.9956	-3.9888	0.00

Table 2: Branch parameters for the 9-bus case. All series and charging parameters are in p.u. on the system base.

Branch	From	To	r	x	b
1	1	4	0.00000	0.05760	0.00000
2	4	5	0.01700	0.09200	0.15800
3	5	6	0.03900	0.17000	0.35800
4	3	6	0.00000	0.05860	0.00000
5	6	7	0.01190	0.10080	0.20900
6	7	8	0.00850	0.07200	0.14900
7	8	2	0.00000	0.06250	0.00000
8	8	9	0.03200	0.16100	0.30600
9	9	4	0.01000	0.08500	0.17600

Topologically, buses 1, 2, and 3 connect generator sources into a meshed transmission network through buses 4, 8, and 6 respectively. The load buses 5, 7, and 9 are embedded in the loop

$$4 \to 5 \to 6 \to 7 \to 8 \to 9 \to 4.$$

This loop is important: real-power injections do not simply flow along a single path. They divide according to the network impedances and the bus voltage-angle profile.

4 Discussion of the Solved Operating Point

The solved generator outputs are

$$P_G^{\text{sol}} = 319.641 \text{ MW},$$
$$Q_G^{\text{sol}} = 22.840 \text{ MVAr}.$$

Since the total real load is 315.000 MW, the real-power loss is

$$P_{\text{loss}} = 319.641 - 315.000 = 4.641 \text{ MW}.$$

The slack bus supplies 71.641 MW, slightly less than the case-side scheduled value of 72.300 MW, because the PV generator dispatch and the network losses determine the final balance. Buses 2 and 3 hold their scheduled real-power outputs of 163.000 MW and 85.000 MW.

The voltage magnitudes remain close to nominal. The highest solved voltage is the slack-bus voltage, 1.0400 p.u., while the lowest voltage is at bus 9, 0.9956 p.u.. The lowest-voltage bus is also the largest load bus (125 MW and 50 MVAr), which is consistent with the expected voltage drop around the transmission loop. The voltage angles range from $-3.9888°$ at bus 9 to $9.2800°$ at bus 2. In an AC transmission network, active-power transfer is strongly tied to voltage-angle differences, so these angles are not merely reporting variables; they are part of the solved state that explains the branch real-power flows.

Table 3: Solved branch powers. The "from" and "to" quantities follow the fixture branch orientation; negative values mean flow enters that end of the branch.

From	To	P_f MW	Q_f MVAr	P_t MW	Q_t MVAr	P_ℓ MW	Q_ℓ MVAr
1	4	71.641	27.046	-71.641	-23.923	0.000	3.123
4	5	30.704	1.030	-30.537	-16.543	0.166	-15.513
5	6	-59.463	-13.457	60.817	-18.075	1.354	-31.531
3	6	85.000	-10.860	-85.000	14.955	0.000	4.096
6	7	24.183	3.120	-24.095	-24.296	0.088	-21.176
7	8	-75.905	-10.704	76.380	-0.797	0.475	-11.502
8	2	-163.000	9.178	163.000	6.654	0.000	15.832
8	9	86.620	-8.381	-84.320	-11.313	2.300	-19.694
9	4	-40.680	-38.687	40.937	22.893	0.258	-15.794

Table 3 also illustrates the importance of sign conventions. For example, on branch 8–2, the fixture orientation is from bus 8 to bus 2, but $P_f = -163.000$ MW and $P_t = 163.000$ MW. The real power therefore flows from bus 2 toward bus 8, not from bus 8 toward bus 2. Similarly, on branch 7–8, the negative P_f indicates that real power enters the branch at bus 8 and leaves at bus 7.

Summing $Q_f + Q_t$ over all branches gives approximately -92.160 MVAr. This is not a violation of reactive-power conservation; it reflects the shunt charging susceptance in the line models. Several branches inject more reactive power through their charging elements than they absorb in series reactive drop, so the net branch reactive "loss" can be negative. This is one of the reasons AC power flow must model reactive power explicitly rather than treating it as a small correction to a real-power-only calculation.

5 Takeaways

1. The IEEE 9-bus problem is a minimal but meaningful AC power-flow benchmark: it includes slack, PV, and PQ bus behavior in a small meshed network.
2. The unknown state consists of voltage angles at non-slack buses and voltage magnitudes at PQ buses. Generator reactive powers and slack-bus power are outputs, not fixed inputs.
3. The solved real-power generation exceeds total real load by 4.641 MW, which is the network real-power loss at this operating point.
4. The branch-flow signs must be interpreted relative to the branch orientation. Negative "from" power means the actual real-power flow is opposite to the stored from-to direction.
5. Reactive-power behavior is qualitatively different from real-power loss accounting because line charging can inject reactive power.